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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: EET203

Course Name: MEASUREMENTS AND INSTRUMENTATION

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

- | | | |
|----|--|-----|
| 1 | Define the following terms in measurement
i) Accuracy ii) Resolution iii) Precision | (3) |
| 2 | What are the different standards of measurements? | (3) |
| 3 | Explain the working of Hall effect multipliers. | (3) |
| 4 | What is a TOD meter? | (3) |
| 5 | Derive the equation for capacitance using Schering bridge. | (3) |
| 6 | How high voltage is tested using sphere gaps? | (3) |
| 7 | Write short notes on RTD. | (3) |
| 8 | Describe anyone method of permeability measurement. | (3) |
| 9 | Explain the working of electromagnetic flow meter. | (3) |
| 10 | Explain any three types of oscilloscope probes. | (3) |

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

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|----|-----|--|-----|
| 11 | (a) | Explain with neat diagram the construction and principle of operation of a PMMC instrument. Derive the expression for deflection. | (7) |
| | (b) | Derive the expression for deflection for spring controlled attraction type moving iron instrument. Also explain the type of damping provided in moving iron instruments. | (7) |
| 12 | (a) | What is the different torques needed for proper operation of an indicating instrument? | (8) |

- (b) The coil of a measuring instrument has a resistance of 1Ω and the instrument has a full scale deflection of 250V when a resistance of 4999Ω is connected in series with it. Find (6)
- I. The current range of the instrument when used as an ammeter with the coil connected across a shunt of $1/499\Omega$.
 - II. The shunt resistance for the instrument to give a full scale deflection of 50 A.

Module 2

- 13 (a) With neat diagram explain the construction and working of an electro-dynamometer wattmeter. (8)
- (b) Derive the expression for torque of a single phase induction type energy meter. (6)
- 14 (a) Derive the expression for ratio and phase angle error in a current transformer. (10)
- (b) Explain how power can be measured in a 3-phase circuit using two wattmeters with a neat circuit diagram in unbalanced load condition. (4)

Module 3

- 15 (a) With the help of diagram explain the working of Kelvin double bridge. (7)
- (b) Draw the circuit and phasor diagram of Maxwell's Inductance bridge and derive the expression for unknown inductance. (7)
- 16 (a) Explain the working of a Meggar with the help of a neat diagram. (6)
- (b) What is a DC potentiometer? Explain the calibration of ammeter and voltmeter using it with neat diagrams. (8)

Module 4

- 17 (a) Explain the method of flux measurement using ballistic galvanometer. (6)
- (b) Describe the construction and working of photoconductive and photovoltaic cells. (8)
- 18 (a) Explain in detail the measurement of iron losses in a magnetic material using Lloyd-Fisher square wattmeter method. (8)
- (b) Explain the temperature measurement using thermocouples. (6)

Module 5

- 19 (a) Explain the basic principle and working of LVDT. (6)
- (b) Write short notes on (8)
- I. Digital Multi Meter and
 - II. Clamp on meter
- 20 (a) How strain is measured using a strain gauge? (4)
- (b) Draw a neat block diagram of a Cathode Ray Oscilloscope and describe the function of each block in detail. (10)
